

15/06/26

DBMS practice.

Normal Forms.

1NF → Table should not contain any Multivalued Attribute.

Roll no	Name	Course
1	Sai	C/C++
2	Harsh	Java
3	Omkar	C/DBMS

PK.

Rollno	Name
1	Sai
2	Harsh
3	Omkar

Method 3

PK → Roll no.

Method 1

Roll no.	Name.	Course
1	Sai	C
1	Sai	C++
2	Harsh	Java
3	Omkar	C
3	Omkar	DBMS

fk

Roll no.	Course
1	C
1	C++
2	Java
3	C
3	DBMS

PK-combined.

Method 2.

Roll no.	Name	Course 1	Course 2
1	Sai	C	C++
2	Harsh	Java	Null
3	Omkar	C	DBMS

2NF →

- Must be in 1NF
- All the Non-prime attributes of the table should be fully functional on PK (candidate key).

Attributes which are not participate in the format of PK.

Customer

Customer ID	Store ID	Location
1	1	Delhi
1	3	Mumbai
2	1	Delhi
3	2	Bangalore
4	3	Mumbai

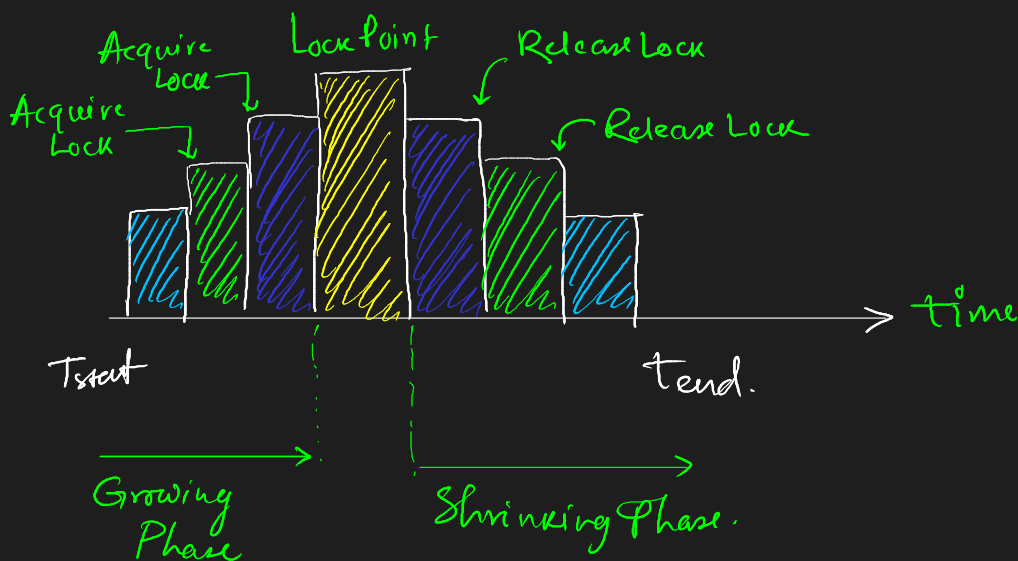
Store ID	Location
1	Delhi
2	Bangalore
3	Mumbai

Customer ID	Store
1	1
1	3
2	1
3	2
4	3

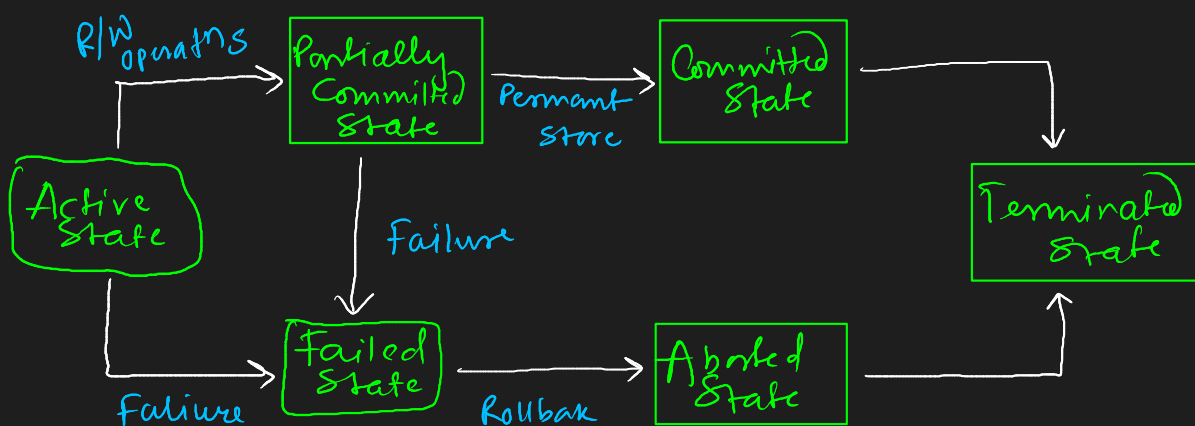
ck: customer id, store id
 prime attributes: customer id
 store id

Not prime: Location.

2 phase Locking Protocol



Transition States.



Q. $S_1 = w_1(A) w_2(A) r_3(B) w_3(B) r_1(B)$

$S_2 = \{r_1(x); r_2(y); r_3(y); w_2(y); w_1(x); w_3(x); r_2(x); w_2(x)\}$

Transactions

$T_1: w_1(A), r_1(B)$

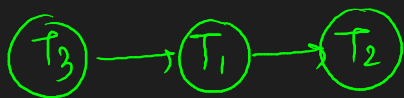
$T_2: w_2(A)$

$T_3: w_3(B), r_3(B)$

Conflicting Operations

A: $w_1(A) \rightarrow w_2(A)$
(Write-Write)
Creates edge $T_1 \rightarrow T_2$

B: $w_3(B) \rightarrow r_1(B)$
(Write-Read)
Creates edge $T_3 \rightarrow T_1$



Conflict serializable = Yes.

Transactions

$T_1: r_1(x), w_1(x)$

$T_2: r_2(y), w_2(y), r_2(x), w_2(x)$

$T_3: r_3(y), w_3(x)$

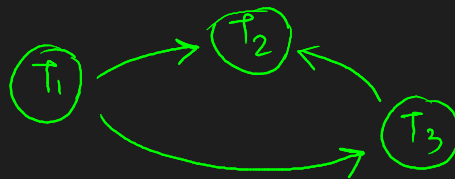
Conflicting operations

x: $r_1(x) \rightarrow w_3(x)$ (r-w) $T_1 \rightarrow T_3$

$r_1(x) \rightarrow w_2(x)$ (r-w) $T_1 \rightarrow T_2$

$w_3(x) \rightarrow r_2(x)$ (w-r) $T_3 \rightarrow T_2$

y: $r_3(y) \rightarrow w_2(y)$ (r-w) $T_3 \rightarrow T_2$



Conflict serializable = Yes

B+ tree 2, 3, 5, 11, 17, 19, 23, 29, 31

• max children = 4

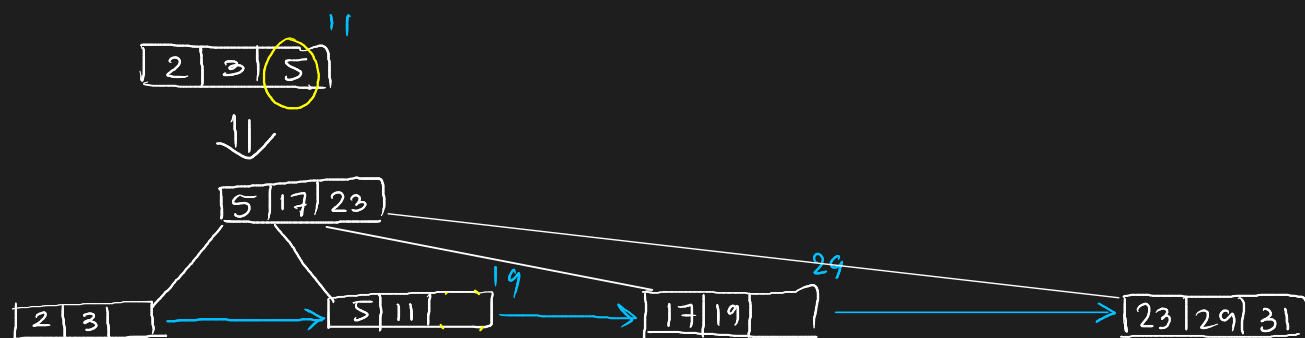
• min children = $\left\lceil \frac{4}{2} \right\rceil = 2$

• min keys = $\left\lceil \frac{m}{2} \right\rceil - 1 = 1$

• max key = 3

(4 pointers, 3 keys)

$K = P - 1$



Q. 3 records or 10 key pointers.

$$\begin{aligned} \text{Data File block} &\Rightarrow \left\lfloor \frac{n}{3} \right\rfloor \\ \text{Data index block} &\Rightarrow \left\lfloor \frac{n}{10} \right\rfloor \end{aligned} \rightarrow Am = \left\lfloor \frac{n}{3} \right\rfloor + \left\lfloor \frac{n}{10} \right\rfloor$$

Q. total block size $(B) = 1 \text{ kb} = 1024 \text{ b}$

data record pointer size $(R_p) = 7 \text{ b}$

Value/Search key field size $(V) = 9 \text{ b}$

block ptr. size $(P_b) = 6 \text{ b}$.

$m = \text{no. of pairs.}$

$$(m \times 16) + 6 \leq 1024$$

$$16m + 6 \leq 1024$$

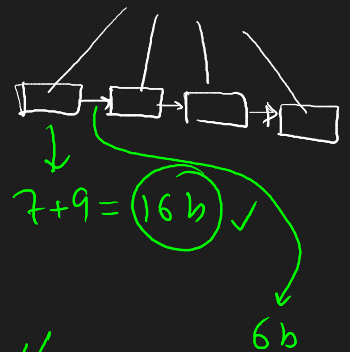
$$8m + 3 \leq 512$$

$$8m \leq 509$$

$$m \leq 63.625$$

$$\therefore m = 63$$

$\therefore \text{order of leaf node} = 63 \checkmark$



$R(ABCDEF)$ $\xrightarrow{\text{Partial dummy.}}$

FD: $\{C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B\}$

CK: $EC = FADB$

$EC^+ = ECFADB$

Partial dependency $\left\{ \begin{array}{l} \text{LHS should be} \\ \text{proper subset of CK} \\ \text{\& RHS should be a} \\ \text{Non-prime attr.} \end{array} \right.$

1. CK = $\{EC\}$

2. Prime Attr = $\{E, C\}$

3. Non prime Attr. = $\{A, B, D, F\}$

$\therefore \text{Not in 2NF.}$

3NF

• Must be in 2NF

• There should be no Transitive dependency in Table.

Q. $R(A, B, C, D)$ FD: $AB \rightarrow C, DA \rightarrow C$

CK: $AB^+ = ABCD$

CK = $\{AB\}$

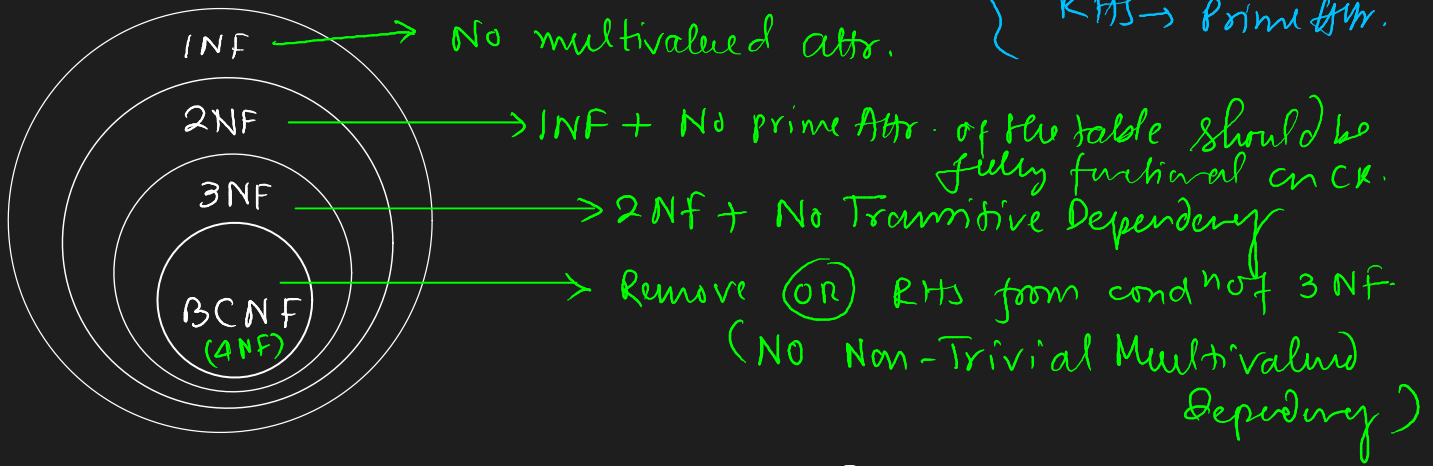
$DA^+ = DABC$

$\therefore \text{CK} = \{AB, DA\}$

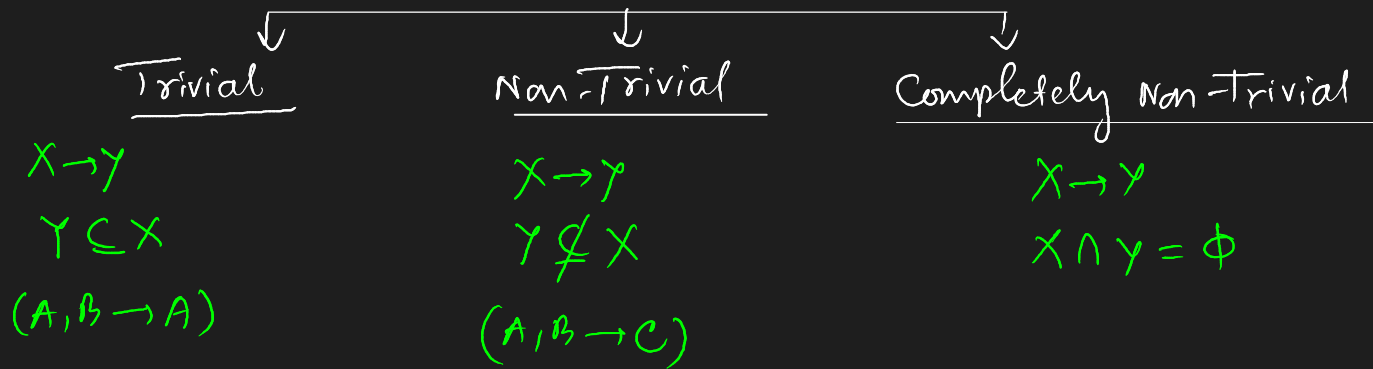
Prime Attr = $\{A, B, D\}$
 Non Prime Attr = $\{C\}$

both valid.
 $\therefore$ 3NF.

LHS $\rightarrow$ CK or SK.
 (OR)
 RHS $\rightarrow$ Prime Attr.



Functional Dependency



$R(AB CDE F)$, $A \rightarrow B$, $C \rightarrow DF$, $AC \rightarrow E$, $D \rightarrow F$

$AC^+ = AC B E D F$ $\cancel{AC} \rightarrow$ so no candidate key $\therefore$ CK = $\{AC\}$

$R(V, W, X, Y, Z)$, $Z \rightarrow Y$, $Y \rightarrow Z$, $X \rightarrow Y$, $X \rightarrow V$, $VW \rightarrow X$

$xy^+ = xyvz$ ✓

$R(A, B, C, D, E)$, $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$,
 $G = \{A \rightarrow CD, E \rightarrow AH\}$

check F covers G.

$A^+ = ACD \Rightarrow (A \rightarrow CD) \text{ in } G$

$E^+ = EADCH \Rightarrow (E \rightarrow AH) \text{ in } G$

check G covers F

$A^+ = ACD \Rightarrow (A \rightarrow C) \text{ in } F$

$E^+ = EAHCD \Rightarrow (E \rightarrow AD) \text{ in } F$

$\therefore F \equiv G$ ✓

MVD



$A \twoheadrightarrow B$

for a single value $A \rightarrow$ then exist multiple value of B

3 level arch

3 Scheme Arch

